IDEATION PHASE

Empathize, Define & Prioritize

Global Fuel Prices and Taxation Analysis (2020–2026)

Project purpose: To learn the concepts of Tableau and data analytics.

STUDENT / PROJECT FIELD	DETAILS
Student Name	Disha Patidar
Enrollment Number	en25ca3020057
Course / Branch	BCA — Computer Applications
Year / Semester	Second Year, 3rd Semester
Institution	MEDI-CAPS University
Project Type	Individual Project
Project Title	Global Fuel Prices and Taxation Analysis (2020–2026)
Project Period Covered	2020–2026 (as stated in project title)

Purpose of this document

This ideation document frames the project around the information needs of people who compare fuel prices, taxation, subsidies, regions, and economic groups. It records project-defined user personas, problem statements, candidate analysis ideas, and a reasoned prioritization for the Tableau work.

> **Scope note:** The personas and problem statements below are analytical design aids. They are not presented as findings from interviews or field research.

1. Empathy Map Canvas

An empathy map organizes what a target user may notice, think, say or do, feel, struggle with, and hope to achieve. For this project, it is used to design views that make comparisons easier and encourage careful interpretation.

How the empathy map is applied

DIMENSION	DESIGN QUESTION FOR THIS PROJECT
Says	What questions might the user ask about fuel prices, tax rates, subsidies, and differences between countries?
Thinks	What explanations or comparisons might the user be considering, and what evidence would help them assess those ideas?
Does	Which comparisons, rankings, filters, and historical views would help the user explore the dataset?
Feels	Where might the user feel uncertain, overloaded by detail, or cautious about drawing conclusions?
Pains	Which aspects of large country-level tables or changing values make comparisons difficult?
Gains	What clearer patterns, concise summaries, and interactive exploration would make the analysis useful?

Design principles carried forward

- Make country and regional comparisons readable.
- Keep units, labels, and category names clear and consistent.
- Use ranking and filtering to reduce visual overload.
- Show descriptive relationships without claiming that one variable causes another.
- Make historical patterns distinguishable from cross-sectional comparisons.

2. Project Personas

The following three personas represent likely analytical user types for the project's three main dashboard scenarios. They are constructed for design planning rather than claimed as real interview participants.

Persona 1 — Energy Policy Consultant

Scenario: Fuel Taxation & Subsidy Policy

EMPATHY DIMENSION	PROJECT-SPECIFIC VIEW
Thinks & feels	Needs to compare policy patterns across countries, while recognizing that fuel prices can be influenced by multiple factors.
Sees	Country-level tax percentages, petrol-price differences by region, subsidy categories, and variation between countries.
Says & does	Compares tax rates with petrol prices and explores how prices differ across subsidy categories.
Pains	Large tables make country comparisons slow; a visible relationship between tax and price does not by itself establish causation.
Gains	Readable comparisons, visible regional patterns, and interactive views that support policy exploration.

Persona 2 — Global Supply Chain Manager

Scenario: Global Fuel Prices & Supply Chain Planning

EMPATHY DIMENSION	PROJECT-SPECIFIC VIEW
Thinks & feels	Wants to understand relative fuel-cost differences and historical movement to support early-stage planning.
Sees	Petrol and diesel prices across countries and regions, high- and low-price country groups, and historical Brent crude and fuel-price trends.
Says & does	Reviews ranked country comparisons, compares petrol with diesel, and examines changes over time.
Pains	Price variation across countries and periods makes quick comparisons difficult.
Gains	Ranked views, regional summaries, and historical context to inform further procurement or logistics analysis.

2. Project Personas (continued)

Persona 3 — Energy Economics Researcher

Scenario: Fuel Prices, Taxation & Economic Development

EMPATHY DIMENSION	PROJECT-SPECIFIC VIEW
Thinks & feels	Wants to explore how prices and tax levels vary across income groups and how subsidy categories relate to observed price patterns.
Sees	Tax-versus-price patterns, prices by subsidy level, average petrol price by income group, and average tax percentage by income group.
Says & does	Filters by income level, compares economic groups, and explores relationships among taxes, subsidies, and prices.
Pains	Group averages can hide country-level variation, and complex relationships require careful interpretation.
Gains	Comparable group-level views and focused filters that provide a clear starting point for further economic analysis.

Implications for dashboard design

The dashboards should make comparisons easy to scan, maintain consistent units and labels, and provide filters that help users focus on relevant subsets. Any patterns should be described as associations or descriptive differences unless the analysis includes a suitable causal method.

> **Evidence boundary:** This document does not claim that the personas were interviewed or that their views were empirically measured.

3. Define the Problem Statements

A problem statement expresses a target user's goal, the barrier they encounter, the reason the barrier exists, and the impact on their work. The statements below translate the three dashboard scenarios into practical design needs.

ID	I AM…	I'M TRYING TO…	BUT…	BECAUSE…	WHICH MAKES ME…
PS-1	An energy policy consultant.	Compare tax percentages, petrol prices, and subsidy patterns across countries.	It is difficult to scan many countries and measures together.	The data contains multiple countries, regions, categories, and periods; tax-price association is not proof of cause.	Want a clearer, interactive way to inspect the evidence.
PS-2	A global supply chain manager.	Identify relatively high- and low-price countries, compare petrol and diesel, and review historical trends.	Useful comparisons take time to find.	Prices vary by geography and time, and the volume of information can overwhelm a quick review.	Need a consolidated visual overview to support further planning.
PS-3	An energy economics researcher.	Compare average prices and tax rates across income groups and subsidy categories.	Group-level patterns may be difficult to interpret.	Country-level differences can be hidden by averages, and several variables may interact.	Need a clear starting point for careful economic exploration.

Problem-statement summary

The central design challenge is to turn multi-country fuel-price and taxation data into readable, interactive comparisons that help users explore policy, supply-chain, and economic-development questions without overinterpreting descriptive patterns.

4. Brainstorm & Idea Listing

This stage lists candidate analytical ideas before prioritization. Because this is an individual project, the ideas are recorded as the student's planning options rather than as outputs of a team brainstorming session.

NO.	IDEA / ANALYSIS	CATEGORY	USER NEED ADDRESSED
1	Compare fuel-tax percentages with average petrol prices across countries.	Taxation & policy	Supports policy comparison and exploration of tax-price patterns.
2	Compare average petrol prices by region and identify relatively high- and low-price countries.	Regional comparison	Makes geographic variation easier to scan.
3	Explore fuel-price patterns across subsidy categories and income groups.	Subsidy & economic development	Supports group-level economic exploration.
4	Compare petrol and diesel prices and review historical fuel-price trends alongside Brent crude.	Time series & supply chain	Adds product and historical context to planning.
5	Provide interactive filters for country, region, income group, or subsidy category where supported by the data.	Exploration & usability	Lets users narrow the view to a relevant comparison.
6	Use clear titles, consistent units, readable legends, and concise labels across dashboards.	Dashboard usability	Reduces ambiguity and cognitive load.

Idea selection criteria

Candidate ideas are assessed using practical feasibility, relevance to the stated user problem, clarity of interpretation, and fit with the three dashboard scenarios. The selected work should be possible with the available dataset and Tableau features.

5. Idea Prioritization

The following ranking is a project-planning judgment. It is not a measured score from user testing. The first three items map directly to the three main dashboard scenarios; the historical trend view acts as supporting analysis.

PRIORITY	FINAL IDEA	FEASIBILITY	IMPORTANCE	SELECTED	REASON
1	Fuel taxation and subsidy policy dashboard	High	High	Yes	Directly addresses comparisons of tax, price, and subsidy patterns.
2	Global fuel prices and supply-chain comparison dashboard	High	High	Yes	Makes country and regional price variation easier to review.
3	Fuel prices, taxation, and economic development dashboard	High	High	Yes	Enables comparison across income groups and subsidy categories.
4	Historical fuel-price and Brent crude trend worksheet	High	Medium	Yes	Provides supporting time context for price analysis.
5	Consistent labels, legends, and interactive filtering	High	High	Yes	Improves readability and helps users focus their comparisons.

Planned dashboard set

DASHBOARD	PRIMARY QUESTION
Fuel Taxation & Subsidy Policy	How do fuel tax rates and prices vary across countries and subsidy categories?
Global Fuel Prices & Supply Chain Planning	Which countries or regions show relatively higher or lower fuel prices, and how do values change over time?
Fuel Prices, Taxation & Economic Development	How do average fuel prices and tax rates differ across income groups?

Project links for reference

Tableau Story: https://public.tableau.com/app/profile/disha.patidar7671/viz/GlobalFuelPricesandTaxationAnalysis2020-2026/STORY
Website: https://global-fuel-prices-taxation-analysis.onrender.com/
GitHub: https://github.com/DishaPatidar11/global-fuel-prices-taxation-analysis

6. Ideation Phase Outcome

The ideation work translates the project topic into three user-oriented analysis scenarios and a prioritized set of dashboard requirements.

Key decisions

• Focus the project on fuel prices, taxation, subsidies, regional differences, income groups, and historical context.
• Organize the core Tableau work around three dashboards: policy, supply-chain planning, and economic development.
• Retain a historical fuel-price/Brent crude trend as supporting analysis.
• Favor readable comparisons, clear labels, consistent units, and useful filtering.
• Interpret charts as descriptive evidence; do not imply causation from correlation alone.

Expected value

The resulting dashboards are intended to make multi-country fuel data easier to explore, help users identify patterns for further investigation, and provide a structured visual starting point for policy, planning, and economic questions.

Limitations and responsible interpretation

The value of each comparison depends on the completeness, definitions, and consistency of the source data. Differences between countries may reflect factors beyond taxation or subsidies. Group averages may conceal country-level variation. Findings should therefore be framed as observed patterns, not causal conclusions.

Prepared for: Disha Patidar · MEDI-CAPS University
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